

The Doppler operator: a generation calculus, closed forms, and exact Sunyaev–Zeldovich signals

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July 2026

Abstract

The boost-operator approach to Compton scattering (Chluba & Ravenni, 2026; Chluba & Rosenberg, 2026; Rosenberg & Chluba, 2026; Hoey et al., 2026) reduces frequency integrals to the action of a differential operator $\hat{O}_\nu = -\nu\partial_\nu$, but the physically assembled object — the *Doppler operator* $\hat{\mathcal{D}}_{\ell\ell'\ell''}^m(\nu, \beta) = \gamma^{-1-1}\hat{\mathcal{B}}(\nu, -\beta)^0\hat{\mathcal{B}}(\nu, \beta)$, the round trip into and out of the electron rest frame — was introduced in Rosenberg & Chluba (2026) with its coefficient structure left as future work. We supply that structure. We derive and machine-verify a complete generation calculus for $\hat{\mathcal{D}}$: a reflection symmetry $\hat{\mathcal{D}}(\hat{O}_\nu) = \hat{\mathcal{D}}^\top(3 - \hat{O}_\nu)$ that forces all diagonal elements to be polynomials in $\hat{\mathcal{D}}_\nu = \hat{O}_\nu^2 - 3\hat{O}_\nu$; a closed first-order rapidity ODE that generates all Taylor tables without ever multiplying two boost operators, together with an all-index companion flow; raising relations for each of the three indices; and closed forms for every element as finite combinations of $J_w = 2\sinh(w\eta)/w$ — shown, via a rationalization lemma, to hold for *every* spin weight and azimuthal number. Structurally, the calculus is representation theory of $SL(2, \mathbb{C})$: the reflection is self-adjointness on photon phase space, boosts are celestial dilations, and on the spectral line $q = -\frac{3}{2} + ik$ the exact single-momentum Thomson matrix is Hermitian, its marginal mode is the boosted comoving equilibrium, and its spectrum is exactly k -independent ($T(k) = W_k T(0) W_k^{-1}$ with a known unitary W_k): the single-momentum Compton dispersion relation is flat, and all genuine dispersion is thermal. As applications we (i) prove the polarized identity of Rosenberg & Chluba (2026, Eq. 20) that its authors “were unable to explicitly prove”, by an exchange symmetry ${}_s\mathcal{K}^{d,m} = {}_m\mathcal{K}^{d,s}$ and an on-curve swap lemma; (ii) assemble the exact thermal SZ operator in closed form (elementary + Bessel + one irremovable transcendent with a complete calculus); (iii) assemble the frame-exact kinematic corrections, recovering the classical C_1 of Nozawa et al. (1998) symbolically and extending it two orders; and (iv) release machine-readable exact tables. Every identity is verified by an open harness suite (30 scripts, ~150 checks, typical residuals 10^{-20} – 10^{-27} ; key results proven symbolically).

1 Introduction

Relativistic corrections to the Sunyaev–Zeldovich (SZ) effect and to CMB spectral distortions are classically computed by momentum-space integrals of the Compton collision term (Nozawa et al., 1998; Chluba et al., 2012). The boost-operator approach (Chluba & Ravenni, 2026) replaces these integrals with operator algebra: an aberration kernel ${}^d\mathcal{K}_{\ell\ell'}^m(\beta)$ acting in multipole space, promoted to a *boost operator* ${}^d\hat{\mathcal{B}}(\nu, \beta) = \mathcal{K}^{d+\hat{O}_\nu}(-\beta)$ whose Doppler weight is shifted by the frequency-derivative operator $\hat{O}_\nu = -\nu\partial_\nu$. Chluba & Rosenberg (2026) used this to obtain the exact thermal SZ operator, and Rosenberg & Chluba (2026) extended the method to polarization, introducing the physically assembled round trip

$$\hat{\mathcal{D}}_{\ell\ell'\ell''}^m(\nu, \beta) = \frac{1}{\gamma} {}^{-1}\hat{\mathcal{B}}_{\ell\ell'}(\nu, -\beta) {}^0\hat{\mathcal{B}}_{\ell'\ell''}(\nu, \beta), \quad (1)$$

the *Doppler operator*: transform to the scatterer’s frame, scatter through the channel ℓ' , transform back. Its matrix elements carry the entire frequency structure of Thomson-limit scattering, but its coefficient structure — the analog of the kernel recursions of Dai & Chluba (2014); Chluba & Ravenni (2026) — was left open.

*calin.cristian@gmail.com. All results are machine-verified; the harness suite and interactive catalog are available at <https://adriencr19.github.io/doppler-operator/>.

This paper closes that program. Section 2 presents the core generation calculus; Section 3 the closed forms and their universality; Section 4 the structural theory (self-adjointness, thermal function class, the $SL(2, \mathbb{C})$ dictionary, the proof of Rosenberg & Chluba 2026, Eq. 20, and the Compton dispersion relation); Section 5 the applications (polarized sector, kinematic corrections, the exact thermal SZ operator, data products); Section 6 the verification methodology. Throughout, q denotes the power-law index of a test spectrum ν^q , on which $\hat{O}_\nu \nu^q = -q\nu^q$ reduces every operator statement to matrix algebra (Lemma L1 below); all statements hold at the operator level by polynomial identity in q .

Conventions

We use the kernel ${}^d\mathcal{K}_{\ell\ell'}^m(-\beta) = 2\pi \int \bar{Y}_{\ell m}(\mu') \bar{Y}_{\ell' m}(\frac{\mu'+\beta}{1+\beta\mu'}) [\gamma(1+\beta\mu')]^{-d} d\mu'$, transposition ${}^d\mathcal{K}_{\ell\ell'}(-\beta) = {}^{2-d}\mathcal{K}_{\ell'\ell}(+\beta)$ (weight and argument flip; verified convention), parity ${}_s\mathcal{K}(-\beta) = (-1)^{\ell+\ell'} {}_s\mathcal{K}(+\beta)$, and $C_\ell^m = \sqrt{(\ell^2 - m^2)/(4\ell^2 - 1)}$ (at $m = 0$: $C_\ell = \ell/\sqrt{4\ell^2 - 1}$). Rapidity $\eta = \text{artanh } \beta$, $\gamma = \cosh \eta$, $p = \sinh \eta$.

Lemma 1 (power-law reduction, L1). *On ν^q , $\hat{O}_\nu \nu^q = -q\nu^q$, so $\hat{\mathcal{D}}_{\ell\ell'\ell''}^m(q) = \gamma^{-1} \mathcal{K}_{\ell\ell'}^{-1-q,m}(+\beta) \mathcal{K}_{\ell'\ell''}^{-q,m}(-\beta)$: every operator identity below is equivalent to a one-parameter family of scalar identities.*

2 The core generation calculus

2.1 Reflection and parity

Theorem 1 (N1, reflection). $\hat{\mathcal{D}}_{\ell\ell'\ell''}^m(q) = \hat{\mathcal{D}}_{\ell''\ell'\ell}^m(-3-q)$; equivalently $\hat{\mathcal{D}}(\hat{O}_\nu) = \hat{\mathcal{D}}^\top(3 - \hat{O}_\nu)$.

Corollary 1 (N1a). *Every outer-diagonal element ($\ell = \ell''$) is invariant under $q \rightarrow -3-q$ and hence a polynomial in $\hat{\mathcal{D}}_\nu = \hat{O}_\nu^2 - 3\hat{O}_\nu$ order by order in β . This proves the empirical observation of Rosenberg & Chluba (2026, §2.3).*

Parity (N2): $\hat{\mathcal{D}}(-\beta) = (-1)^{\ell+\ell''} \hat{\mathcal{D}}(\beta)$. The reflection is not an accident of the Thomson limit; it is self-adjointness of the boost on photon phase space (Section 4).

2.2 The rapidity ODE: generating tables without multiplying boosts

Theorem 2 (N3). *At fixed intermediate index ℓ' ,*

$$\begin{aligned} \partial_\eta \hat{\mathcal{D}}_{\ell\ell'\ell''}^m = & -(\ell + 3 - \hat{O}_\nu) C_{\ell+1}^m \hat{\mathcal{D}}_{\ell+1,\ell',\ell''}^m + (\ell - 2 + \hat{O}_\nu) C_\ell^m \hat{\mathcal{D}}_{\ell-1,\ell',\ell''}^m \\ & - (\ell'' + \hat{O}_\nu) C_{\ell''+1}^m \hat{\mathcal{D}}_{\ell,\ell',\ell''+1}^m + (\ell'' + 1 - \hat{O}_\nu) C_{\ell''}^m \hat{\mathcal{D}}_{\ell,\ell',\ell''-1}^m - \beta \hat{\mathcal{D}}_{\ell\ell'\ell''}^m, \end{aligned} \quad (2)$$

with seed $\hat{\mathcal{D}}|_{\eta=0} = \delta_{\ell\ell'} \delta_{\ell'\ell''}$: a closed first-order system whose iteration generates every Taylor table of $\hat{\mathcal{D}}$ without ever multiplying two boost operators. It is manifestly covariant under N1.

An all-index companion (N3') moves ℓ' as well, at the cost of the off-diagonal family $X_{\ell;a,b;\ell''}$ (middle slots allowed to differ): the outer shifts and the middle shifts each equal $\partial_\eta \hat{\mathcal{D}} + \beta \hat{\mathcal{D}}$ (the exact *cross-slot balance*), and their mean is the combined flow. The same first-order structure underlies the middle-index recursion (N10) and the pure-physical relations (P0–P4, below): a first-order ∂_η acts on one scattering leg at a time, so any ℓ' -move is inherently off-diagonal, while the physical operators close on the outer indices.

2.3 Raising each index

Outer indices are raised by N5/N6 (a two-term ladder with weight-lattice companions ${}^{(j,k)}\hat{\mathcal{D}} = \gamma^{-1} {}^{-1-j}\hat{\mathcal{B}}(-\beta) {}^{-k}\hat{\mathcal{B}}(\beta)$, which collapse to a single label by the gauge identity ${}^{(j+1,k+1)}X(q) = {}^{(j,k)}X(q+1)$); the middle index by N10 on the off-diagonal family; and the *pure-physical* scheme (P0–P4, obtained in reconciliation with an independent derivation) removes all auxiliary objects from the outer recursions: axis ladders seeded by $\hat{\mathcal{D}}_{000}$, the outer-pair relation, and the rank-one factorization $\hat{\mathcal{D}}_{h\ell\ell'} = \hat{\mathcal{D}}_{h\ell'0} \hat{\mathcal{D}}_{0\ell\ell'}/\hat{\mathcal{D}}_{0\ell'0}$. A sum rule (N4) and the seed catalog (N7, with a ${}_2F_1$ closed form for $\hat{\mathcal{D}}_{mmm}^m$) complete the generator set.

3 Closed forms: the J -calculus and its universality

3.1 The bracket table

All $m = 0$ kernel columns needed by the SZ system reduce to the family $J_w = 2 \sinh(w\eta)/w$ (with $J_0 = 2\eta$, $J_{-w} = J_w$, $E_w = 2 \cosh(w\eta)$) and the collapse rule $J_{a+q}J_{b+q} = [E_{a+b+2q} - E_{a-b}]/[(a+q)(b+q)]$. Writing $\mathcal{K}_{ab}(q) \equiv {}^{-1-q}\mathcal{K}_{ab}(+\beta)$, the Doppler elements are *bracket products*; e.g.

$$\hat{\mathcal{D}}_{000}(q) = \frac{J_{2+q}J_{1+q}}{4\gamma p^2}, \quad \hat{\mathcal{D}}_{101}(q) = \frac{3}{4\gamma p^4}(\gamma J_{2+q} - J_{3+q})(\gamma J_{1+q} - J_q), \quad (3)$$

and similarly $\hat{\mathcal{D}}_{020}, \hat{\mathcal{D}}_{202}, \hat{\mathcal{D}}_{121}, \hat{\mathcal{D}}_{100}, \hat{\mathcal{D}}_{120}$, etc. These reproduce the published Taylor tables and the SZ operator of [Chluba & Rosenberg \(2026, Eq. 11\)](#) symbolically, and extend it (e.g. the p^6 term $\frac{11}{3150}\hat{\mathcal{O}}_\nu(\hat{\mathcal{O}}_\nu^5 - 9\hat{\mathcal{O}}_\nu^4 + 13\hat{\mathcal{O}}_\nu^3 + 57\hat{\mathcal{O}}_\nu^2 - 86\hat{\mathcal{O}}_\nu - 120)$, i.e. $\frac{11}{3150}(\hat{\mathcal{D}}_\nu^3 - 14\hat{\mathcal{D}}_\nu^2 + 40\hat{\mathcal{D}}_\nu)$ by N1a).

3.2 The rationalization lemma: all spins, all m

Lemma 2 (N22/N23). *In ${}_s\mathcal{K}_{\ell\ell'}^{d,m}$ both harmonics carry the same (s, m) , and on the aberration curve $\sin \frac{\theta_+}{2} = \sin \frac{\theta}{2} \sqrt{\frac{1-\beta}{1+\beta\mu}}$, $\cos \frac{\theta_+}{2} = \cos \frac{\theta}{2} \sqrt{\frac{1+\beta}{1+\beta\mu}}$, so the half-angle factors of the product pair up rationally:*

$${}_sY_{\ell m}(\mu) {}_sY_{\ell' m}(\mu_+) = (1-\beta)^{\frac{|m+s|}{2}}(1+\beta)^{\frac{|m-s|}{2}} \frac{\text{poly}(\mu)}{(1+\beta\mu)^{\max(|m|, |s|)}}. \quad (4)$$

With $t = 1 + \beta\mu$ and $\gamma(1 \pm \beta) = e^{\pm\eta}$, $\int t^{n-d} dt = \gamma^{d-n-1} J_{n+1-d}$: every kernel element, any spin weight and azimuthal number, is a finite J -combination.

Consequences: the polarized ($s = 2$) columns and all polarized Doppler elements in closed form; and at $s = 0$, $m \geq 1$ the prefactor is $(1 - \beta^2)^{|m|/2} = \gamma^{-|m|}$, so general- m elements introduce *no new transcendentals* — the m -seed ${}_2F_1$'s are J -reducible at integer arguments. This is the closed-form face of the exchange symmetry of Section 4.3.

4 Structure theory

4.1 Self-adjointness and the thermal function class

On $L^2(\nu^2 d\nu) \times L^2(S^2)$ one has $\hat{\mathcal{O}}_\nu^\dagger = 3 - \hat{\mathcal{O}}_\nu$, and the kernel transposition $d \rightarrow 2 - d$ is the conformal shadow on the celestial sphere: N1 is self-adjointness of the physically assembled round trip (N12). Thermal averages of the closed forms over the relativistic Maxwellian belong to a rigid class (N9): each channel is an iterated Bessel antiderivative, mapped by $(\partial_z^2 - 1)^{\sigma/2}$ (with $\sigma = \ell + 2\ell' + \ell''$) onto finite combinations $\sum_j c_j K_{a_j}(z)$; only the $\sigma = 0$ channel $\langle \hat{\mathcal{D}}_{00} \rangle$ is Bessel-closed. The transcendentals carry a complete recurrence calculus (N11), and the channel structure is organized by a Henyey–Greenstein generating identity (N13, Funk–Hecke). Recoil corrections enter as rungs on the weight lattice (N14), where per-rung reciprocity is broken exactly as elastic scattering requires.

4.2 The $SL(2, \mathbb{C})$ dictionary

Boosts act on the celestial sphere as dilations $\tan \frac{\theta'}{2} = e^{-\eta} \tan \frac{\theta}{2}$; Doppler weight and spin are conformal dimensions $(\Delta, J) = (d, s)$; transposition is the shadow $\Delta \rightarrow 2 - \Delta$. The boost operator is unitary on the weight-matched measure $\nu^{1-2d} d\nu$ for every d (the $d = 1$ kernel is literally orthogonal), and on the unitary Mellin line every weight sees the same principal-series representation $\Delta = 1 - ik$ (N18).

4.3 The proof of RC26 Eq. 20: exchange symmetry and the on-curve swap lemma

[Rosenberg & Chluba \(2026\)](#) observed numerically that the two mixed-spin blocks of the polarized SZ system coincide (${}_{0,\pm 2}\hat{\mathcal{S}}_\ell \equiv {}_{\pm 2,0}\hat{\mathcal{S}}_\ell$, their Eq. 20) and noted they were unable to prove it. The identity is *collective*: it

fails for each azimuthal m separately (so no leg-by-leg kernel identity can prove it) and holds only m -summed. Two results settle it (N19):

Lemma 3 (exchange symmetry). ${}_s Y_{\ell m} = (-1)^{s+m} {}_m Y_{\ell s}$ pointwise (from $d_{-m,s}^\ell = d_{-s,m}^\ell$), hence ${}_s \mathcal{K}_{\ell\ell'}^m(\beta) = {}_m \mathcal{K}_{\ell\ell'}^s(\beta)$ exactly: the kernel cannot distinguish its bundle label from its magnetic label, and m -sums over spin pairs are spin-sums.

Lemma 4 (on-curve swap). The collective four-harmonic sum $G(x, y) = \sum_m {}_{s_1} Y_{\ell m}(x) {}_{s_1} Y_{\ell' m}(gx) {}_{s_2} Y_{\ell m}(y) {}_{s_2} Y_{\ell' m}(gy)$ is symmetric in $x \leftrightarrow y$ pointwise whenever both leg pairs lie on the graph of the same aberration map g . In half-angle coordinates ($g: t \mapsto \lambda t$) this is a rational identity, proved by machine certificate for every instance needed (all $\ell \leq 8$ at $\ell' = 2$; spot checks beyond); off the curve it is false, with a radical obstruction that rationalizes exactly on the curve.

Theorem 3 (N19). $\sum_m {}_{s_1} \mathcal{K}_{\ell\ell'}^{a,m}(\beta) {}_{s_2} \mathcal{K}_{\ell\ell'}^{c,m}(\beta)$ is symmetric under $s_1 \leftrightarrow s_2$ for arbitrary weights a, c ; [Rosenberg & Chluba \(2026, Eq. 20\)](#) is the case $(a, c) = (-1 - q, 2 + q)$, $\ell' = 2$, transposed. It therefore holds to all orders in β , and survives any independent spectral filtering of the two legs.

4.4 The Compton dispersion relation

On the spectral line $q = -\frac{3}{2} + ik$ the exact single-momentum Thomson matrix $T(k) = \hat{\mathcal{D}}_{\ell 0 \ell'} + \frac{1}{10} \hat{\mathcal{D}}_{\ell 2 \ell'} - [\delta - \beta \hat{\mu}]$ is Hermitian (N18). Its two open questions are answered by (N20):

Theorem 4 (marginal mode). $T(k)v = 0$ for $v_\ell = 2\pi \int \bar{Y}_{\ell 0}(\mu) [\gamma(1 - \beta\mu)]^q d\mu$: the zero mode is the boosted comoving equilibrium — the electron’s own rest-frame-isotropic power law, invariant because Thomson scattering is elastic and gain equals loss for any isotropic rest-frame field.

Theorem 5 (exact flatness). $T(k) = W_k T(0) W_k^{-1}$ with $W_k =$ multiplication by $[\gamma(1 - \beta\mu)]^{ik}$ (unitary). Elasticity in the rest frame makes the evolution commute with $(\nu')^{ik}$; the ν^{ik} factor shifts the Mellin block and the angular remainder is W_k . Hence the single-momentum damping spectrum is exactly k -independent, the flat zero band is the W -orbit of the comoving equilibrium, and multiple scattering solves once at the self-dual point: $e^{\tau T(k)} = W_k e^{\tau T(0)} W_k^{-1}$ with $T(0)$ real symmetric.

All genuine k -dependence is thermal: Maxwell–Jüttner averaging preserves Hermiticity but breaks the (p -dependent) intertwining, opening a spectral gap $g(k)$ — the exact per-mode thermalization rate (at 10 keV: $-0.0062, -0.0076, -0.0170, -0.0471$ at $k = 0, 0.7, 2, 4$), far below the angular branches ($\approx -0.77, -0.85$): comptonization’s two-timescale structure as an exact spectral statement.

5 Applications

5.1 The polarized sector

The spin-weighted foundations (transposition, parity with s -flip, ℓ -raising and rapidity ODE with their sm -terms) hold at $s = \pm 2$; polarized N1 proves the $\hat{\mathcal{D}}_\nu$ -structure of the polarization tables ([Rosenberg & Chluba, 2026, Eq. 21a](#)); the mixed-spin reflection $T(s_1, s_2)(\hat{\mathcal{O}}_\nu) = T(s_2, s_1)^\top (3 - \hat{\mathcal{O}}_\nu)$ shows the $I \rightarrow P$ and $P \rightarrow I$ blocks are adjoints; and Eq. 20 is settled (Section 4.3). With the rationalization lemma the polarized sector now runs seeds $\rightarrow$ recursions $\rightarrow$ closed forms $\rightarrow$ exact thermal averages, and its exact p^6 tables are released as a data product.

5.2 Frame-exact kinematic corrections

A reflection obstruction (N16) shows the physical kSZ operator $-[10x\partial_x + \frac{47}{5}x^2\partial_x^2 + \frac{7}{5}x^3\partial_x^3]$ ([Nozawa et al., 1998](#)) is *not* of the form $-q f(\hat{\mathcal{D}}_\nu)$, while any assembly of outer-diagonal Doppler elements is: the kinematic signal cannot live on the diagonal. The frame-exact assembly (N21) follows [Chluba & Rosenberg \(2026, §3\)](#): photons isotropic, kinematics in the boosted Maxwellian ($f_\ell \propto I_{\ell+1/2}$) and the flux factor $\tau^* = \tau/(1 - \beta_p \mu_p)$, and scattering through the outer off-diagonal *monopole columns* $\hat{S}_\ell = \sqrt{2\ell+1}[\hat{\mathcal{D}}_{\ell 00} + \frac{1}{10}\hat{\mathcal{D}}_{\ell 20}] - \delta_{\ell 0} - \beta \delta_{\ell 1}$. Our generator reproduces [Chluba & Rosenberg \(2026, Eq. 28\)](#) at p^3, p^5 and Nozawa’s C_1 symbolically; the

$\tau^* \rightarrow \tau$ shift is proved to be exactly $+\hat{D}_\nu$ (the classical $\beta_p \mu_p Y_0$ term); the next orders are generated in both conventions,

$$C_2^* = -\left[25 x \partial_x + \frac{1117}{10} x^2 \partial_x^2 + \frac{847}{10} x^3 \partial_x^3 + \frac{183}{10} x^4 \partial_x^4 + \frac{11}{10} x^5 \partial_x^5\right], \quad (5)$$

$$C_3^* = -\left[\frac{75}{4} x \partial_x + \frac{21873}{40} x^2 \partial_x^2 + \frac{49161}{40} x^3 \partial_x^3 + \frac{27519}{35} x^4 \partial_x^4 + \frac{6684}{35} x^5 \partial_x^5 + \frac{3917}{210} x^6 \partial_x^6 + \frac{64}{105} x^7 \partial_x^7\right] \quad (6)$$

(C_3^* beyond the printed literature), and the closed forms make the kinematic dipole exact by a single quadrature at any temperature.

5.3 The exact thermal SZ operator

$\langle \hat{D}_{02} \rangle$ provably cannot share the Bessel-closed form of $\langle \hat{D}_{00} \rangle$ (closure/rigidity, N9); the optimal assembly is

$$\hat{S}_{\text{th}} = \frac{K_{3-2\hat{O}_\nu}(z) - K_1(z)}{2\theta_e(2 - \hat{O}_\nu)(1 - \hat{O}_\nu)K_2(z)} + \frac{z}{10K_2(z)} \sum_{j=1}^8 c_j(\hat{O}_\nu) F(-2, a_j, z) - 1, \quad z = 1/\theta_e, \quad (7)$$

elementary + Bessel + exactly one irremovable transcendent $F(\mu, a, z) = \int e^{-z \cosh \eta} \cosh(a\eta) \sinh^{2\mu} \eta d\eta$ with a complete calculus (N11) and three exact evaluation routes. Quantitatively, at 50 keV the p^8 tower errs by $\sim 10\%$ and is *worse* than p^4 (the series is genuinely asymptotic), while the assembly is exact.

5.4 Data products

Machine-readable exact tables: the $\mathcal{O}(p^8)$ scalar tables (501 entries, channels $\ell' \leq 3$, outer ≤ 6 , exact rational q -polynomial coefficients) and the polarized $\mathcal{O}(p^6)$ tables ($T(2, 2), T(0, 2), T(2, 0)$; $m = 0, 1, 2$; $\ell, \ell'' \leq 4$), both generated from the ODE/closed forms and spot-verified against quadrature.

6 Verification

Every identity in this paper is verified by an open Wolfram-language harness suite (30 scripts): high-precision quadrature of the defining integrals (25–40 digits) on power-law spectra at non-integer q , multiple speeds and azimuthal numbers, against every derived recursion, closed form, ODE (via symbolic-step finite differences), and the published tables. Key results are established symbolically (exact zeros): the closed forms vs. the published tables, the kinematic C_1 and the τ^*/τ shift, the obstruction and gap factorizations, and the on-curve swap certificates (rational identities). Scripts, logs, and per-script pages: <https://adriencr19.github.io/doppler-operator/verification/>.

7 Conclusions

The Doppler operator now has the complete calculus its introduction in [Rosenberg & Chluba \(2026\)](#) called for: symmetries that fix its structure (reflection = self-adjointness; parity; exchange), generators that produce it (rapidity flows; per-index raisings; pure-physical ladders), closed forms universal in spin and azimuthal number, a rigid thermal function class, and a representation-theoretic frame in which its deepest properties (unitarity, the flat single-momentum dispersion relation, the comoving marginal mode) are one-line statements. The applications — the settled polarized identity, the frame-exact kinematic tower, the exact thermal operator — indicate that the practical SZ program can now be run exactly, with series expansions demoted to conveniences. Natural continuations are the recoil sector (Kompaneets-type equations via the weight lattice, cf. [Hoey et al. 2026](#)), multiple scattering with the exact exponentiation of Section 4.4, and the uniform general- ℓ proof of the on-curve swap lemma.

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